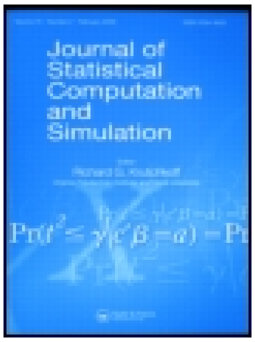

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Bayesian analysis of the *p*-order integer-valued AR process with zero-inflated Poisson innovations

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ABSTRACT

In recent years, there has been a considerable interest to study count time series with a dependence structure and appearance of excess of zeros values. Such series are commonly encountered in diverse disciplines, such as economics, financial research, environmental science, public health, among others. In this paper, we propose a stationary *p*-order integer-valued autoregressive process with zero-inflated Poisson innovations, called the ZINAR(*p*) times series model. We study some of its theoretical properties and develop a Markov chain Monte Carlo (MCMC) algorithm for inferring parameters from Bayesian perspectives. Finally, we demonstrate the utility of proposed ZINAR(*p*) model through simulated and real data examples.

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1. Introduction

A popular approach to deal with count time series having a structure depending on their past observations, is to consider the integer-valued autoregressive (INAR) process, as proposed by Al-Osh and Alzaid [1] and McKenzie [2], in which the process is established by considering a Poisson distribution for the innovations. However, in some cases, the Poisson distribution may be inappropriate because of the property of equal mean and variance. In this regard, Ristić et al. [3] proposed a geometric first-order integer-valued autoregressive process with geometric marginals based on a negative binomial thinning operator. For count data, overdispersion is not uncommon because the occurrence of zero counts might be greater than expected under the Poisson assumption. In this case, a zero-inflated Poisson distribution should be more suitable to model the innovations. The zero-inflated Poisson model has been widely used in many disciplines, including manufacturing applications [4], medicine [5], public health [6], environmental sciences [7], agriculture [8–10], among others. To handle with an excess of zeros in count time series, Jazi et al. [11] introduced the stationary ZINAR(1) process with innovations having a zero-inflated Poisson distribution. The authors also showed that the marginal distribution of the process is

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zero-inflated. Afterward, Barreto-Souza [12] defined a zero-modified geometric INAR(1) process, which is able to capture deflation or inflation of zeros as well as overdispersion and underdispersion.

There some related works on count time series, but most of them focused on the first-order case. To analyse a counting process with a more flexible model, Alzaid and Al-Osh [13] presented an integer-valued p-th order autoregressive INAR(p) model and investigated some aspects of the behaviour of the process. Also, Neal and Subba Rao [14] considered a wide class of integer-valued autoregressive moving-average (INARMA) processes, in which the estimation is carried out by performing the MCMC method for Bayesian inference.

To the best of our knowledge, there are no studies related to the analysis of data sets with excess of zeros and dependence structure with order larger than one, in both frequentist and Bayesian approaches. To deal with this situation we propose an extension of the INAR(p) model, with zero-inflated Poisson innovations, called the ZINAR(p) model, for the analysis of count series with excess of zeros and time dependence with order larger than one. A Bayesian approach is conducted by developing an efficient MCMC algorithm to proceed the estimation of the parameters in the model. The performance is demonstrated by simulated and an original real data sets. It is important to emphasize that, the inference from the Bayesian perspective may result in richer inferences than does the likelihood-based approach in the case of small samples.

The paper is organized as follows. Section 2 provides some preliminaries concerning the INAR process. Section 3 outlines the proposed ZINAR(p) model and discusses the stationarity condition along with some of its mathematical properties, including the likelihood function as well. Section 4 presents an MCMC algorithm for Bayesian posterior and predictive inferences. The methodology developed in the paper are illustrated in Section 5 with simulated and real data. Finally, Section 6 concludes with short remarks and some possible directions for future research.

2. Preliminaries

One method of dealing with count time series is to consider a p-order integer-valued autoregressive (INAR(p)) process, which is commonly employed to model non-negative integer-valued phenomena. Suppose that X is a non-negative integer-valued random variable and $\alpha \in [0, 1]$. The '$\circ$' operator [15] is defined as

$$\alpha \circ Y = \sum_{i=1}^{Y} V_i, \tag{1}$$

where V_i are independent and identically distributed (i.i.d.) random variables, independent of Y, which follows a Bernoulli distribution with parameter α. Thus, if $y > 0$,

$$\alpha \circ Y | Y = y \sim \text{Bin}\left(y; \alpha\right), \tag{2}$$

where $\text{Bin}(y; \alpha)$ represents the Binomial distribution with parameters $y \in \{1, 2, \ldots\}$ and $\alpha \in [0, 1]$. If $y = 0$, $\alpha \circ Y | Y = y$ follows a degenerated distribution at zero.

Note that

$$E\left[\alpha \circ Y\right] = E\left\{E\left[\alpha \circ Y | Y\right]\right\} = \alpha E\left[Y\right]. \tag{3}$$

Definition 2.1 ([16]): The INAR(p) process is defined by

$$Y_t = \sum_{i=1}^{p} \alpha_i \circ Y_{t-i} + Z_t, \quad t \in \mathbb{Z}, \tag{4}$$

where $\alpha_i \in [0,1]$ $i = 1, \dots, p$ and $\{Z_t\}$ is an i.i.d. sequence of non-negative integer-valued random variables, called *innovations*. As commented by Jin-Guan and Yuan [16], this supposition for $\{Z_t\}$ makes the innovations independent of $Y_{t-1}, \dots, Y_{t-p}$ and it is stronger than the assumption made by Al-Osh and Alzaid [1], which define $\{Z_t\}$ as a sequence of uncorrelated non-negative integer-valued random variables. Beside this, the following assumptions are assumed:

(1) For all t, Z_t is independent of all the random variables V_i in (1).
(2) Z_t has mean μ and finite variance σ^2, for all t.
(3) All random variables V_i are mutually independent.

As inflation of zeros frequently occurs when modelling counting data, it is necessary to consider models that capture this nature of the data, which is the case of the ZINAR(1) process proposed by Jazi et al. [11]. In this work, the innovations Z_t are supposed to be distributed as a ZIP(ρ, λ) distribution, with probability mass function (pmf)

$$\mathbb{P}(Z_t = k) = \rho \mathbb{I}_{\{0\}}(k) + (1-\rho)\frac{e^{-\lambda}\lambda^k}{k!}, \quad k = 0, 1, 2, \dots, \tag{5}$$

where $\lambda > 0$, $0 \leq \rho \leq 1$, and $\mathbb{I}_B(\cdot)$ denotes the indicator function of the set B. As commented before, these authors show that the marginal process has an approximate ZIP distribution, but the intractability of the marginal distribution leads to a complicated maximum-likelihood-based estimation method. As a consequence, the authors did not consider interval estimation and the asymptotic distribution of the estimators. In order to deal with excess of zeros and to propose a more flexible model, we extend the work of Jazi et al. [11] by considering a p-order integer-valued autoregressive process with ZIP innovations, denoted by ZINAR(p) process, from a Bayesian point of view.

3. The ZINAR(p) process

3.1. Description and stationarity condition

The ZINAR(p) process is an integer-valued p-order autoregressive time series with ZIP innovations, as defined by (4) with $Z_t \sim$ ZIP(ρ, λ). As the ZINAR(p) process is a na-tural extension of the INAR(p) process, we can utilize some important results about the stationarity condition for the INAR(p) process. In particular, Jin-Guan and Yuan [16] showed that the stationarity conditions of INAR(p) and AR(p) are the same, as stated in the following theorem:

Theorem 3.1: *Let $\{Z_t\}_{t\in\mathbb{Z}}$ be i.i.d. non-negative integer-valued random variables with $E[Z_t] = \mu$, $Var[Z_t] = \sigma^2 < \infty$ and suppose that $\alpha_i \in [0,1]$, for $i = 1, \dots, p$. If the solutions of*

$$B^p - \alpha_1 B^{p-1} - \cdots - \alpha_{p-1} B - \alpha_p = 0$$

are inside the unit circle, then there exists a unique stationary non-negative integer-valued random series $\{Y_t\}$ *satisfying*

$$Y_t = \alpha_1 \circ Y_{t-1} + \cdots + \alpha_p \circ Y_{t-p} + Z_t;$$
$$Cov\,(Y_s, Z_t) = 0 \quad \text{for } s < t.$$

Proof: See Theorem 2.1 in Jin-Guan and Yuan [16]. ■

Another important result about the stationarity condition of the INAR(p) process was obtained by Doukhan et al. [17]. The authors showed that for discrete valued processes the weak dependence implies mixing conditions under natural assumptions, and further verified that the $\sum_{i=1}^{p} \alpha_i < 1$ is the standard condition for stationarity and ergodicity of the INAR(p) process. Since the ZINAR(p) process have the same stochastic structure as the INAR(p) process, we can use $\sum_{i=1}^{p} \alpha_i < 1$ to ensure the stationarity of the ZINAR(p) process.

3.2. Some mathematical properties

In the following theorem, we present some mathematical properties about the ZINAR(p) process, including the mean and autocovariance function.

Theorem 3.2: *Let* $\{Y_t\}$ *be a stationary ZINAR(p) process with innovations* $Z_t \sim \text{ZIP}(\rho, \lambda)$. *The mean* μ *and the covariance function* $\gamma(\cdot)$ *of* $\{Y_t\}$ *are, respectively, given by*

$$\mu = \frac{\lambda\,(1-\rho)}{1 - \sum_{i=1}^{p} \alpha_i} \tag{6}$$

and

$$\gamma\,(h) = \sum_{i=1}^{p} \alpha_i \gamma\,(h-i) + \mathbb{I}_{\{0\}}(h) \lambda\,(1-\rho)\,(1+\rho\lambda)\,. \tag{7}$$

Proof: From (5), we have that $E[Z_t] = (1-\rho)\lambda$ and $Var[Z_t] = \lambda(1-\rho)(1+\rho\lambda)$. Thus,

$$\begin{aligned}
\mu = E\,[Y_t] &= E\left\{E\left[Y_t | Y_{t-1}, \ldots, Y_{t-p}\right]\right\} \\
&= E\left\{E\left[\sum_{i=1}^{p} \alpha_i \circ Y_{t-i} + Z_t \,\middle|\, Y_{t-1}, \ldots, Y_{t-p}\right]\right\} \\
&= \sum_{i=1}^{p} \alpha_i E\,[Y_{t-i}] + E\,[Z_t] \qquad (8) \\
&= \sum_{i=1}^{p} \alpha_i \mu + \lambda\,(1-\rho)\,, \qquad (9)
\end{aligned}$$

where we used (3) and the fact that Z_t is not correlated with $Y_{t-1}, \ldots, Y_{t-p}$ to obtain Equation (8), and the stationarity of Y_t to obtain (9). Then, Equation (6) follows immediately.

For the covariance function, we have

$$\begin{aligned}\gamma(h) = Cov\left[Y_t, Y_{t-h}\right] &= Cov\left[\sum_{i=1}^{p} \alpha_i \circ Y_{t-i} + Z_t, Y_{t-h}\right] \\ &= Cov\left[\sum_{i=1}^{p} \alpha_i \circ Y_{t-i}, Y_{t-h}\right] + Cov\left[Z_t, Y_{t-h}\right] \\ &= \sum_{i=1}^{p} Cov\left[\alpha_i \circ Y_{t-i}, Y_{t-h}\right] + Cov\left[Z_t, Y_{t-h}\right]. \end{aligned} \tag{10}$$

For $h \geq p$, the first term of (10) equals

$$\begin{aligned} Cov\left[\alpha_i \circ Y_{t-i}, Y_{t-h}\right] &= E\left[\left(\alpha_i \circ Y_{t-i}\right) Y_{t-h}\right] - E\left[\alpha_i \circ Y_{t-i}\right] E\left[Y_{t-h}\right] \\ &= E\left[E\left[\left(\alpha_i \circ Y_{t-i}\right) Y_{t-h} | Y_{t-1}, \ldots, Y_{t-h}\right]\right] \\ &- \quad E\left[E\left[\alpha_i \circ Y_{t-i} | Y_{t-1}, \ldots, Y_{t-h}\right]\right] E\left[Y_{t-h}\right] \\ &= \alpha_i \left\{E\left[Y_{t-i} Y_{t-h}\right] - E\left[Y_{t-i}\right] E\left[Y_{t-h}\right]\right\} \\ &= \alpha_i \gamma\left(h - i\right). \end{aligned} \tag{11}$$

For the second part of (10), there exists a nonzero correlation between Z_t and Z_{t-h} if and only if $h = 0$, that is

$$Cov\left[Z_t, Z_{t-h}\right] = \begin{cases} 0, & h \neq 0, \\ \lambda\left(1 - \rho\right)\left(1 + \lambda\rho\right), & h = 0. \end{cases} \tag{12}$$

■

3.3. The likelihood function

In what follows, for any random vectors $\mathbf{X}$ and $\mathbf{Y}$, we use $\pi(\mathbf{x})$ to denote the density function of $\mathbf{X}$ and $\pi(\mathbf{x}|\mathbf{y})$ to denote the conditional density function of $\mathbf{X}|\mathbf{Y} = \mathbf{y}$.

Let $Y_1, \ldots, Y_{n+p}$, $n > p$, be samples from the ZINAR(p) process. Denote the vector of all unknown parameters by $\boldsymbol{\theta} = (\boldsymbol{\alpha}, \rho, \lambda)$, where $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \ldots, \alpha_p)$. We further denote the joint pmf of $Y_1, \ldots, Y_{n+p}$ given $\boldsymbol{\theta}$ by $\pi(y_1, \ldots, y_{n+p}|\boldsymbol{\theta})$. Then, the likelihood function can be written as

$$\pi\left(\mathbf{y}|\boldsymbol{\theta}\right) = \pi\left(y_1, \ldots, y_p|\boldsymbol{\theta}\right) \prod_{t=p+1}^{p+n} \pi\left(y_t | Y_{t-1} = y_{t-1}, \ldots, Y_{t-p} = y_{t-p}, \boldsymbol{\theta}\right), \tag{13}$$

where

$$\begin{aligned} &\pi\left(y_t | Y_{t-1} = y_{t-1}, \ldots, Y_{t-p} = y_{t-p}, \boldsymbol{\theta}\right) \\ &= \sum_{k_1=0}^{\min\{y_{t-1}, y_t\}} \binom{y_{t-1}}{k_1} \alpha_1^{k_1} \left(1 - \alpha_1\right)^{y_{t-1} - k_1} \end{aligned}$$

$$\times \sum_{k_2=0}^{\min\{y_{t-2}, y_t - k_1\}} \binom{y_{t-2}}{k_2} \alpha_2^{k_2} (1-\alpha_2)^{y_{t-2}-k_2}$$

$$\times \quad \vdots$$

$$\times \sum_{k_p=0}^{\min\{y_{t-p}, y_t - k_1 \cdots - k_{p-1}\}} \binom{y_{t-p}}{k_p} \alpha_p^{k_p} (1-\alpha_p)^{y_{t-p}-k_p}$$

$$\times \left[\rho \mathbb{I}_{\{0\}}(y_t - k_1 \cdots - k_p) + (1-\rho) \frac{e^{-\lambda} \lambda^{y_t - k_1 \cdots - k_p}}{[y_t - k_1 \cdots - k_p]!} \right]. \tag{14}$$

The exact likelihood function given by (13) is intractable because the likelihood due to the p initial values $\pi(y_1, \ldots, y_p|\boldsymbol{\theta})$ is unknown. An usual approach to estimate the parameters is to consider the so-called conditional analysis, where $\pi(y_1, \ldots, y_p|\boldsymbol{\theta})$ is ignored. More details refer to Hamilton [18], Bu et al. [19], Prado and West [20, Sec 2.3.5] and Jazi et al. [11].

4. Bayesian inference for the ZINAR(p) process

4.1. Prior distributions

To complete the Bayesian treatment of the ZINAR(p) process, we need to specify prior distributions for the unknown parameters in the model. Assuming that the prior distributions of the components of $\boldsymbol{\theta}$ are independent *a priori*, we have

$$\pi(\boldsymbol{\theta}) = \pi(\boldsymbol{\alpha}, \rho, \lambda) = \pi(\rho)\, \pi(\lambda) \prod_{i=1}^{p} \pi(\alpha_i).$$

In the absence of good prior information from historical data or from previous experiment, we use a convenient strategy of considering diffuse proper priors in order to avoid improper posterior distributions. Thus, we consider the following setup:

$$\pi(\alpha_i) \sim \mathrm{U}(0;1); \quad \text{for } i = 1, \ldots, p, \text{ satisfying } \sum_{i=1}^{p} \alpha_i < 1;$$

$$\pi(\rho) \sim \mathrm{U}(0;1);$$

$$\pi(\lambda) \sim \mathrm{Gamma}(A_\lambda; B_\lambda), \quad \text{with hyperparameters } A_\lambda \text{ and } B_\lambda \text{ being positive,}$$

where $\mathrm{U}(a;b)$ denotes the uniform distribution with parameters $-\infty < a < b < \infty$, and $\mathrm{Gamma}(c;d)$ denotes the gamma distribution with parameters $c > 0$, $d > 0$ and mean c/d.

As suggested by Lin and Lee [21], Garay et al. [22] and Garay et al. [9], the values of the hyperparameters are kept fixed and chosen relatively small for ensuring flatness.

4.2. Bayesian estimation and predictive inference

The MCMC algorithm

In the Bayesian framework, it is well known that an efficient way to approximate characteristics of the posterior distribution, like expectations, median, modes, etc. is through the generation of posterior samples via MCMC methods [23]. In what follows, we develop an MCMC algorithm using a data augmentation scheme by combining the observed data and some latent variables described below. The strategy is similar to that proposed by Neal and Subba Rao [14].

Let $\mathbf{y} = \{y_1, y_2, \dots, y_{n+p}\}$, be a sample from a ZINAR(p) process, with $n > p$. Assuming that the order $p > 0$ is known, we define the following latent variables:

$$S_{t,i} = \alpha_i \circ Y_{t-i}, \quad \text{for } i = 1, \dots, p.$$

Using Equation (2), we have that the distribution of $S_{t,i}|Y_{t-i} = y_{t-i}$, α_i is $\text{Bin}(y_{t-i}; \alpha_i)$ if $y_{t-i} > 0$, and it is a degenerated distribution at zero if $y_{t-i} = 0$. Also, the random variables associated to the distributions of $S_{t,i}|Y_{t-i} = y_{t-i}$, α_i, $i = 1, \dots, p$, are independent. It follows from Equation (5) that the distribution of Z_t is a mixture with components Poiss(λ) and the degenerated distribution at zero, with weights $1 - \rho$ and ρ, respectively. Similar to Ando [24] and Garay et al. [8], there exists a latent binary random variable W_t such that

$$W_t|\rho \sim \text{Bern}(\rho), \tag{15}$$

where Bern(ρ) denotes the Bernoulli distribution with parameter ρ, and

$$\begin{aligned} &Z_t|(W_t = 0, \lambda) \sim \text{Poiss}\,(\lambda)\,, \\ &Z_t|(W_t = 1, \lambda) \quad \text{follows a degenerated distribution at zero.} \end{aligned} \tag{16}$$

Also, from Equation (4), we have that $Z_t = Y_t - \sum_{i=1}^{p} S_{t,i}$. Let $\mathbf{S}_t = (S_{t,1}, S_{t,2}, \dots, S_{t,p})$ and $\mathbf{y}_{(t-p)} = \{y_{t-1}, y_{t-2}, \dots, y_{t-p}\}$, for $t = p+1, \dots, n$, by the basic hypotheses of the process (4), we have that $S_{t,1}, S_{t,2}, \dots, S_{t,p}$ are independent when $\mathbf{y}_{(t-p)}$, $\boldsymbol{\alpha}$, ρ and λ are given; the same is true for (Z_t, W_t) and $\mathbf{S}_t$.

Thus,

$$\pi\left(z_t, \mathbf{s}_t, w_t|\mathbf{y}_{(t-p)}, \boldsymbol{\theta}\right) = \pi\,(z_t|w_t, \lambda)\,\pi\,(w_t|\rho) \prod_{i=1}^{p} \pi(s_{t,i}|y_{t-i}, \alpha_i), \quad t = p+1\ \dots, n+p, \tag{17}$$

where the distributions of $(S_{t,i}|y_{t-i}, \alpha_i)$, $(Z_t|w_t, \lambda)$ and $W_t|\rho$ are given in Equations (16) and (15), respectively.

By Bayes' theorem,

$$\pi\left(z_t, \mathbf{s}_t, w_t|y_t, \mathbf{y}_{(t-p)}, \boldsymbol{\theta}\right) \propto \pi\left(y_t|z_t, \mathbf{s}_t, w_t, \mathbf{y}_{(t-p)}, \boldsymbol{\theta}\right) \pi\left(z_t, \mathbf{s}_t, w_t|\mathbf{y}_{(t-p)}, \boldsymbol{\theta}\right). \tag{18}$$

Using Equation (4) we obtain

$$\pi\left(y_t|z_t, \mathbf{s}_t, w_t, \mathbf{y}_{(t-p)}, \boldsymbol{\theta}\right) = \mathbb{I}_{\{z_t + \sum_{i=1}^{p} s_{t,i}\}}(y_t). \tag{19}$$

Thus, we have that

$$\pi\left(z_t, \mathbf{s}_t, w_t | y_t, \mathbf{y}_{(t-p)}, \boldsymbol{\theta}\right) \propto \pi\left(z_t, \mathbf{s}_t, w_t | \mathbf{y}_{(t-p)}, \boldsymbol{\theta}\right) \mathbb{I}_{A_t}(z_t, \mathbf{s}_t, w_t), \tag{20}$$

where

$$A_t = \left\{(z_t, \mathbf{s}_t, w_t); z_t \in \{0, 1, \ldots\}, 0 \le s_{t,i} \le y_{t-i}, i = 1, \ldots, p, z_t \right.$$
$$\left. + \sum_{i=1}^{p} s_{t,i} = y_t, w_t \in \{0, 1\}\right\}$$

and $\pi(z_t, \mathbf{s}_t, w_t | \mathbf{y}_{(t-p)}, \boldsymbol{\alpha}, \rho, \lambda)$ is obtained through Equation (17).

Now, let us define $\mathbf{S} = (\mathbf{S}_{p+1}, \mathbf{S}_{p+2}, \ldots, \mathbf{S}_{p+n})$, $\mathbf{W} = (W_{p+1}, W_{p+2}, \ldots, W_{p+n})$ and $\mathbf{Z} = (Z_{p+1}, Z_{p+2}, \ldots, Z_{p+n})$. Adopting the prior distribution for $\boldsymbol{\theta}$ given in Section 4.1, the full joint likelihood function of $\mathbf{Z}$, $\mathbf{S}$, $\mathbf{W}$ and $\boldsymbol{\theta}$, given the observed data $\mathbf{y}$ (see Appendix 1) is:

$$\pi\left(\mathbf{z}, \mathbf{s}, \mathbf{w}, \boldsymbol{\theta} | \mathbf{y}\right) \propto \prod_{t=p+1}^{p+n} \pi\left(z_t, \mathbf{s}_t, w_t | \mathbf{y}_{(t-p)}, y_t, \boldsymbol{\theta}\right) \pi\left(\boldsymbol{\theta}\right) \tag{21}$$

$$\propto \prod_{t=p+1}^{p+n} \left\{\rho^{w_t} \left[\frac{e^{-\lambda}\lambda^{z_t}}{z_t!}(1-\rho)\right]^{1-w_t}\right.$$
$$\left. \times \prod_{i=1}^{p} \binom{y_{t-i}}{s_{t,i}} \alpha_i^{s_{t,i}} (1-\alpha_i)^{y_{t-i}-s_{t,i}}\right\} \lambda^{A_\lambda - 1} e^{-B_\lambda \lambda}, \tag{22}$$

when $(z_t, \mathbf{s}_t, w_t) \in A_t$, $t = p+1, \ldots, p+n$ and $\pi(\mathbf{z}, \mathbf{s}, \mathbf{w}, \boldsymbol{\alpha}, \boldsymbol{\theta} | \mathbf{y}) = 0$ otherwise.

Therefore, we can easily obtain the full conditional distributions using (21). We remark that, while the full conditional distributions of the parameters α_i, λ and ρ have standard forms and are easy to sample from, the same is not true for the latent variables Z_t, $S_{t,i}$ and W_t. In this case, we use independence Metropolis-Hastings paths, with proposals generated from the distributions of $S_{t,i}|y_{t-i}$ and $W_t|z_t$, and also taking into account the constraint $Z_t = Y_t - \sum_{i=1}^{p} S_{t,i}$. Below, we present the MCMC algorithm used to proceed the Bayesian posterior inference.

Step 1. Update $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_p)$, one component at a time, either in a fixed or random order. For each $i = 1, \ldots, p$,

(a) Sample a proposed value α_i from $\pi(\alpha_i | \mathbf{z}, \mathbf{s}, \mathbf{w}, \mathbf{y}, \boldsymbol{\alpha}_{[-i]}, \rho, \lambda)$, which is

$$\text{Beta}\left(\sum_{t=p+1}^{p+n} s_{t,i} + 1; \sum_{t=p+1}^{p+n} \left(y_{t-i} - s_{t,i}\right) + 1\right),$$

where $\boldsymbol{\alpha}_{[-i]} = (\alpha_1, \ldots, \alpha_{i-1}, \alpha_{i+1}, \ldots, \alpha_p)$ and $\text{Beta}(\cdot;\cdot)$ denotes the Beta distribution.

(b) If α_i is drawn such that $\sum_{k=1}^{i} \alpha_k < 1$, accept the proposed value α_i. Otherwise, repeat step (a).

Step 2. Draw λ from its full conditional distribution $\pi(\lambda|\mathbf{z},\mathbf{s},\mathbf{w},\mathbf{y},\boldsymbol{\alpha},\rho)$, which is

$$\mathrm{Gamma}\left(\sum_{t=p+1}^{p+n} z_t\,(1-w_t)+A_\lambda;\ \sum_{t=p+1}^{p+n}(1-w_t)+B_\lambda\right).$$

Step 3. Sample ρ from $\pi(\rho|\mathbf{z},\mathbf{s},\mathbf{w},\mathbf{y},\boldsymbol{\alpha},\lambda)$, which is

$$\mathrm{Beta}\left(\sum_{t=p+1}^{p+n} w_t+1;\ \sum_{t=p+1}^{p+n}(1-w_t)+1\right).$$

Step 4. Update the latent vectors $\mathbf{S}$, $\mathbf{Z}$ and $\mathbf{W}$. For $t\in\{p+1,p+2,\dots,p+n\}$, we take samples $(\mathbf{s}_t,z_t,w_t)$ in one block at a time, either in a fixed or random order, via the following independent-proposal Metropolis-Hastings scheme:

(a) For $1\le i\le p$, if $y_{t-i}=0$, then propose $s^*_{ti}=0$ as the value of S_{ti} and go to item (c). Otherwise, go to item (b).
(b) Draw the proposed value s^*_{ti} from the distribution $\mathrm{Bin}(y_{t-i};\alpha_i)$.
(c) If $y_t<\sum_{i=1}^{p}s^*_{ti}$, then repeat item (b).
(d) Set $z^*_t=y_t-\sum_{i=1}^{p}s^*_{ti}$.
(e) Draw w^*_t from the full conditional distribution $\pi(w_t|z^*_t,\mathbf{y},\boldsymbol{\alpha},\rho,\lambda)$, which is given below.
(f) Given the set of samples $\{\mathbf{s}_t^{(j-1)},z_t^{(j-1)},w_t^{(j-1)}\}$ obtained at stage $(j-1)$, the proposed set $\{\mathbf{s}^*_t,z^*_t,w^*_t\}$ is accepted with probability

$$\min\left\{\frac{\pi\left(\mathbf{s}^*_t,z^*_t,w^*_t|\mathbf{y}_{(t-p)},\boldsymbol{\theta}\right)\pi\left(\mathbf{s}_t^{(j-1)}|\mathbf{y}_{(t-p)},\boldsymbol{\alpha}\right)\pi\left(w_t^{(j-1)}|z_t^{(j-1)},\rho,\lambda\right)}{\pi\left(\mathbf{s}_t^{(j-1)},z_t^{(j-1)},w_t^{(j-1)}|\mathbf{y}_{(t-p)},\boldsymbol{\alpha},\boldsymbol{\theta}\right)\pi\left(\mathbf{s}^*_t|\mathbf{y}_{(t-p)},\boldsymbol{\alpha}\right)\pi\left(w^*_t|z^*_t,\rho,\lambda\right)},1\right\},\tag{23}$$

where $\pi(\mathbf{s}_t,z_t,w_t|\mathbf{y}_{(t-p)},\boldsymbol{\theta})$ is given in Equation (17), $\pi(\mathbf{s}_t^{(j-1)}|\mathbf{y}_{(t-p)},\boldsymbol{\alpha})$ is the joint distribution of independent $\mathrm{Bin}(y_{t-i};\alpha_i)$ random variables, $1\le i\le p$, and the conditional distribution of $W_t|z_t,\rho,\lambda$ is given by

$$W_t|Z_t=z_t,\rho,\lambda\sim\mathrm{Bern}(\phi),\quad\text{with }\phi=\frac{\rho}{\rho+(1-\rho)\,\mathrm{e}^{-\lambda}},\ \text{when } z_t=0,$$

and $W_t|Z_t=z_t,\rho,\lambda$ follows a degenerated distribution at zero when $z_t>0$.

Predictive inference

In the analysis of times series data, the prediction of future values has a great impact in many practical applications. Thus, we are often interested in the predictive capabilities of the model. In this sense, we want to obtain the m-step ahead predictive distribution of $\mathbf{Y}_m^{\text{pred}}=(Y_{T+1},Y_{T+2},\dots,Y_{T+m})$, where $T=p+n$. In order to do so, we use a strategy similar to that developed in Neal and Subba Rao [14], by extending the proposed MCMC algorithm to generate samples from the marginal likelihood $\pi(\mathbf{y}_m^{\text{pred}}|\mathbf{y})$. The proposed procedure for predictive inference is stated in Step 5 below

Step 5. Update the values of $\mathbf{Y}_m^{\text{pred}}$ one component at a time, sequentially from y_{T+1} to y_{T+m}, using the following procedure:

(a) Draw the value of w_{T+k} from distribution Bern(ρ).
(b) Draw the value of z_{T+k} from the distribution of $Z_{T+k}|w_{T+k}, \lambda$ given by Equation (16).
(c) For $1 \le i \le p$,

$$s_{T+k,i} = \begin{cases} 0, & \text{if } y_{T+k-i} = 0 \\ \text{Value drawn from Bin}\left(y_{T+k-i}; \alpha_i\right), & \text{otherwise.} \end{cases}$$

(d) Set

$$y_{T+k} = \sum_{i=1}^{p} s_{T+k,i} + z_{T+k}. \tag{24}$$

By proceeding Step 5 of the algorithm, we draw values from the conditional distribution of $\pi(y_{T+k}, z_{T+k}, w_{T+k}, \mathbf{s}_{T+k}|\boldsymbol{\theta}, \mathbf{y}, \mathbf{y}_{k-1}^{\text{pred}})$, $k = 1, \ldots, m$. Then, $\{y_{T+1}, y_{T+2}, \ldots, y_{T+m}\}$ are the desired samples from the posterior marginal likelihood.

5. Numerical examples

We carry out two numerical studies to show the performance of the Bayesian estimation for the ZINAR(p) model. All the computational procedures were implemented using the R software [25]. Codes are available from the authors upon request.

5.1. *Artificial data set*

We consider the ZINAR(p) process with $p \in \{1, 2, 3, 4\}$ and the following parameter setup: $\lambda = 2$, $\rho \in \{0.25, 0.40, 0.70\}$, $\alpha = 0.4$ for the ZINAR(1), $\boldsymbol{\alpha} = (\alpha_1, \alpha_2) = (0.4, 0.2)$ for the ZINAR(2), $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \alpha_3) = (0.2, 0.3, 0.15)$ for the ZINAR(3) and $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \alpha_3, \alpha_4) = (0.3, 0.2, 0.15, 0.1)$ for the ZINAR(4). We generated 1,000 samples of size $n \in \{80, 15, 300\}$ from each fixed ZINAR(p) model, and fitted it to each data set using the proposed MCMC algorithm, obtaining 150,000 MCMC draws from the joint posterior distribution of the parameters. The first 30,000 (20%) iterations were discarded as burn-in samples. To eliminate potential problems due to autocorrelation, the subsequent equally spaced draws with thinning equal to 10 were retained.

The prior distributions are described in Section 4.1. As suggested by Garay et al. [8] and Garay et al. [9], the chosen values of the hyperparameters were $A_\lambda = 0.01$ and $B_\lambda = 0.01$. Each parameter was estimated by its posterior mean, which is approximated by the average across the retained 12,000 MCMC draws. Also, a 95% equal-tailed credibility interval was obtained for each parameter.

Fixing $\rho = 0.25$, Tables 1–3 show the average values (MC Mean) and standard deviations (MC SD) of the MCMC estimates across all 1,000 samples. Also, it is shown the proportion of the 95% credibility intervals containing the true value of the parameter (MC

Table 1. Artificial data sets. Summary of MCMC results based on 1000 simulated samples from the ZINAR(p) process, for $\rho = 0.25$, $n = 80$ and $p = 1, \ldots, 4$.

			MCMC summary		
Model	Parameter	Real value	MC Mean	MC SD	MC Cov
	α	**0.40**	0.3834	0.0842	96.1%
ZINAR(1)	λ	**2.00**	2.0350	0.2949	94.4%
	ρ	**0.25**	0.2405	0.1012	97.9%
	α_1	**0.40**	0.3774	0.0947	96.1%
ZINAR(2)	α_2	**0.20**	0.2175	0.0925	97.1%
	λ	**2.00**	2.0838	0.4222	95.8%
	ρ	**0.25**	0.2648	0.1397	97.9%
	α_1	**0.20**	0.2175	0.0905	96.4%
	α_2	**0.30**	0.2820	0.0926	94.7%
ZINAR(3)	α_3	**0.15**	0.1852	0.0857	94.2%
	λ	**2.00**	1.9823	0.6316	90.5%
	ρ	**0.25**	0.2226	0.1808	98.3%
	α_1	**0.30**	0.2853	0.0881	96.1%
	α_2	**0.20**	0.2123	0.0857	97.4%
ZINAR(4)	α_3	**0.15**	0.1880	0.0811	93.8%
	α_4	**0.10**	0.1567	0.0753	91.5%
	λ	**2.00**	1.9624	1.0689	98.1%
	ρ	**0.25**	0.2185	0.2384	99.2%

Table 2. Artificial data sets. Summary of MCMC results based on 1000 simulated samples from the ZINAR(p) process, for $\rho = 0.25$, $n = 150$ and $p = 1, \ldots, 4$.

			MCMC summary		
Model	Parameter	Real value	MC Mean	MC SD	MC Cov
	α	**0.40**	0.3933	0.0627	96.5%
ZINAR(1)	λ	**2.00**	2.0023	0.2135	95.5%
	ρ	**0.25**	0.2418	0.0803	96.6%
	α_1	**0.40**	0.3894	0.0713	96.6%
ZINAR(2)	α_2	**0.20**	0.2003	0.0727	97.5%
	λ	**2.00**	2.0414	0.3001	96.2%
	ρ	**0.25**	0.2480	0.1121	97.6%
	α_1	**0.20**	0.2037	0.0730	95.3%
	α_2	**0.30**	0.2847	0.0736	95.3%
ZINAR(3)	α_3	**0.15**	0.1645	0.0699	97.3%
	λ	**2.00**	2.0179	0.3878	95.2%
	ρ	**0.25**	0.2450	0.1271	90.8%
	α_1	**0.30**	0.2844	0.0740	95.0%
	α_2	**0.20**	0.1956	0.0736	97.2%
ZINAR(4)	α_3	**0.15**	0.1627	0.0694	96.4%
	α_4	**0.10**	0.1266	0.0628	95.6%
	λ	**2.00**	1.9798	0.5368	92.3%
	ρ	**0.25**	0.2369	0.1703	93.9%

Cov). From these results, we can see that the proposed MCMC algorithm produces accurate and precise estimates in all cases. The results for the cases $\rho = 0.40$ and $\rho = 0.70$ are similar, and are presented in Tables A1–A6 in Appendix 2.

5.2. *Real data set*

As an empirical illustration, we consider the *Assaults dataset* about weekly count of assaults. All occurrences are described, recorded and organized by SSI/UFPE, which is an office

Table 3. Artificial data sets. Summary of MCMC results based on 1000 simulated samples from the ZINAR(p) process, for $\rho = 0.25$, $n = 300$ and $p = 1, \ldots, 4$.

			MCMC summary		
Model	Parameter	Real value	MC Mean	MC SD	MC Cov
	α	**0.40**	0.3934	0.0446	95.2%
ZINAR(1)	λ	**2.00**	2.0184	0.1504	95.2%
	ρ	**0.25**	0.2438	0.0588	96.3%
	α_1	**0.40**	0.3965	0.0513	95.8%
ZINAR(2)	α_2	**0.20**	0.1937	0.0541	97.0%
	λ	**2.00**	2.0393	0.2093	96.6%
	ρ	**0.25**	0.2442	0.0846	95.4%
	α_1	**0.20**	0.2011	0.0547	95.5%
	α_2	**0.30**	0.2903	0.0530	93.0%
ZINAR(3)	α_3	**0.15**	0.1544	0.0531	94.1%
	λ	**2.00**	2.0103	0.2614	94.8%
	ρ	**0.25**	0.2506	0.1054	96.9%
	α_1	**0.30**	0.2948	0.0552	97.0%
	α_2	**0.20**	0.1943	0.0569	95.3%
ZINAR(4)	α_3	**0.15**	0.1509	0.0547	96.1%
	α_4	**0.10**	0.1118	0.0497	96.6%
	λ	**2.00**	2.0275	0.3533	95.1%
	ρ	**0.25**	0.2571	0.1324	98.0%

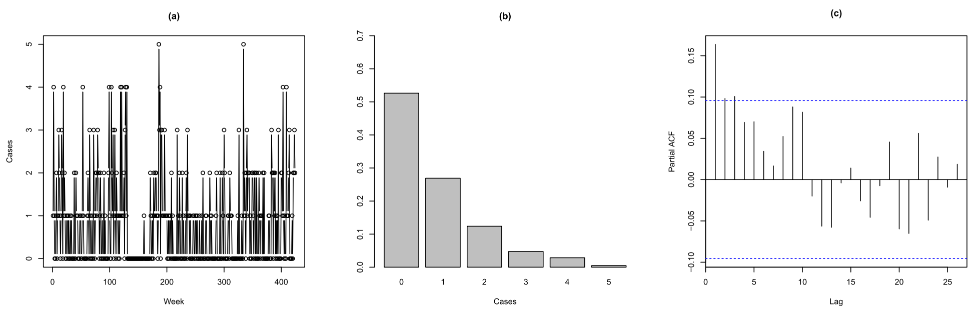

Figure 1. Weekly count of assaults: (a) time evolution, (b) bar plot and (c) PACF plot.

in charge of the planning, execution and evaluation of projects and activities related to institutional security at the Federal University of Pernambuco (UFPE) of Brazil. To the best of our knowledge, this is the first time this data set is analysed in the scientific literature. These data consist of 422 observations obtained during the period from January 2008 to December 2015. The two latest values were preserved for prediction comparison purposes and the remaining were used for estimation and two-step-ahead prediction.

Panels (a) and (b) of Figure 1 depict the time evolution and the bar plot of the weekly count of assaults, respectively. The two plots indicate a significant presence of zero values observations (52%). Furthermore, the partial autocorrelation function (PACF) in panel (c) suggests a structural dependence with order $p > 1$. In our analysis, we consider the INAR(p) and ZINAR(p) processes with $p = 1, 2, 3$ for comparative purposes.

For each model, we generated $L = 150,000$ draws from the posterior distribution, discarded the first 30,000 iterations as a burn-in period and considered a spacing of size 5 to reduce autocorrelation. The convergence of the chains was monitored using trace plots and autocorrelation plots. The final 24,000 realizations from the target posterior distribution

Table 4. Assaults data set. MCMC summaries for the INAR(p) and ZINAR(p) processes, $p = 1, 2, 3$.

		MCMC summaries				
Model	Parameter	Mean	SD	Median	$Q_{0.025}$	$Q_{0.975}$
	α	0.146	0.040	0.145	0.067	0.225
INAR(1)	λ	0.682	0.049	0.681	0.589	0.784
	α_1	0.136	0.040	0.135	0.059	0.217
INAR(2)	α_2	0.089	0.041	0.088	0.019	0.169
	λ	0.612	0.128	0.611	0.513	0.721
	α_1	0.133	0.039	0.133	0.055	0.213
INAR(3)	α_2	0.076	0.036	0.074	0.011	0.152
	α_3	0.096	0.039	0.095	0.022	0.176
	λ	0.551	0.055	0.549	0.447	0.665
	α	0.200	0.041	0.200	0.119	0.279
ZINAR(1)	λ	1.171	0.114	1.168	0.959	1.406
	ρ	0.452	0.055	0.454	0.338	0.554
	α_1	0.179	0.041	0.179	0.098	0.260
ZINAR(2)	α_2	0.104	0.041	0.103	0.026	0.187
	λ	1.155	0.128	1.152	0.916	1.418
	ρ	0.505	0.061	0.508	0.379	0.618
	α_1	0.173	0.042	0.172	0.091	0.256
	α_2	0.088	0.040	0.087	0.015	0.169
ZINAR(3)	α_3	0.085	0.041	0.084	0.012	0.170
	λ	1.138	0.139	1.134	0.878	1.424
	ρ	0.539	0.065	0.542	0.401	0.657

are used for making Bayesian inference. Some summary statistics are shown in Table 4, namely: mean, standard deviation, median, 2.5% and 97.5% quantiles.

We use Bayes factors to compare the models, which involve the calculation of marginal densities of the data under each model. The use of Bayes factors is a general tool to compare models in several contexts. For its application in models with autoregressive structures, see McCabea and Martin [26] and Lin and Lee [21]. Assuming that two given models are equally probable, the Bayes factor is equal to the ratio between the marginal densities under each one.

In this application, we consider the models INAR(1), which will be denoted by M_1, INAR(2) (M_2), INAR(3) (M_3), ZINAR(1) (M_4), ZINAR(2) (M_5) and ZINAR(3) (M_6). The marginal density for each model is given by

$$f\left(\mathbf{y}|M_k\right) = \int f\left(\mathbf{y}|\boldsymbol{\theta}_k, M_k\right) \pi\left(\boldsymbol{\theta}_k|M_k\right)\, \mathrm{d}\boldsymbol{\theta}_k, \quad \text{for } k = 1, \ldots, 6, \tag{25}$$

where $\boldsymbol{\theta}_k$, $f(\mathbf{y}|\boldsymbol{\theta}_k, M_k)$ and $\pi(\boldsymbol{\theta}_k|M_k)$ represent the parameters, the likelihood function and the prior density of $\boldsymbol{\theta}_k$ under the M_k model, respectively. Clearly, this is the expectation of $f(\mathbf{y}|\boldsymbol{\theta}_k, M_k)$ with respect to $\pi(\boldsymbol{\theta}_k|M_k)$. For the ZINAR($p$) models the computation of this integral is a computationally intensive task. A simple way to overcome this issue is to use an approximation based on the MCMC posterior samples. For example, we can use importance sampling with importance function given by the posterior density – see Kass and Raftery [27] for more details. To be specific,

$$f\left(\mathbf{y}|M_k\right) \approx \left\{ \frac{1}{L} \sum_{l=1}^{L} \frac{1}{f\left(\mathbf{y}|\boldsymbol{\theta}_k^{(l)}, M_k\right)} \right\}^{-1}, \tag{26}$$

Table 5. Assaults data set. Estimated marginal densities under several ZINAR(p) models.

Model (M_k)	$f(y\|M_k)$
INAR(1)	$1.8432 \times e^{-225}$
INAR(2)	$1.9062 \times e^{-224}$
INAR(3)	$1.8584 \times e^{-223}$
ZINAR(1)	$1.2712 \times e^{-217}$
ZINAR(2)	$9.3442 \times e^{-217}$
ZINAR(3)	$3.2738 \times e^{-216}$

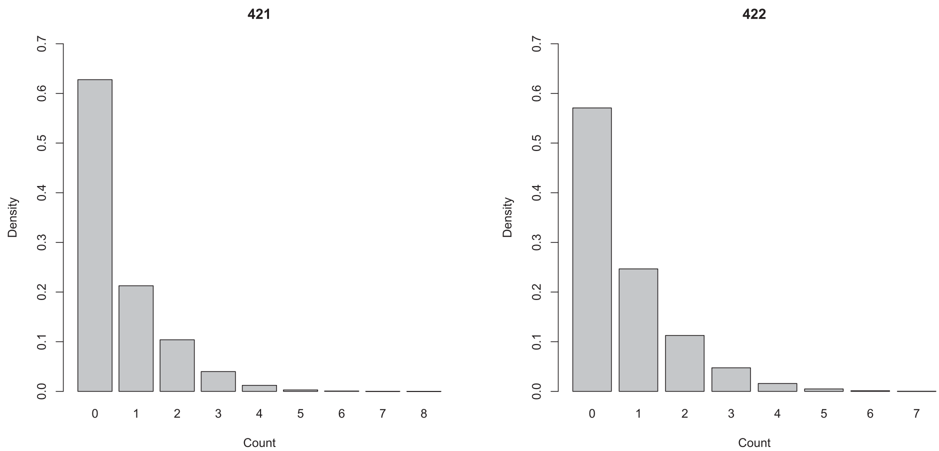

Figure 2. Assaults data set. Bar plot of the predictive distributions of data points **421** and **422**, under ZINAR(3) model.

where $\boldsymbol{\theta}_k^{(l)}$ is the draw of the posterior distribution of $\boldsymbol{\theta}_k$ obtained at the l—th iteration of the MCMC algorithm. The results are presented in Table 5.

The Bayes factors of the ZINAR(2) model relative to INAR(1), INAR(2), INAR(3), ZINAR(1) and ZINAR(3) models are approximately 5.06950×10^8, 4.9020040×10^7, 5.028089×10^6, 7.350434 and 0.28542, respectively. Thus, it appears that the ZINAR(2) model is more appropriated than the INAR(1), INAR(2), INAR(3) and ZINAR(1) models. In addition, the Bayes factor favours the ZINAR(3) model when compared to the others.

Under the ZINAR(3) model, by the definition of the ZIP distribution, we have that an estimate of the proportion of zeros $\mathbb{P}(Z_t = 0)$ is given by

$$\frac{1}{L}\sum_{l=1}^{L}\left\{\widehat{\rho}^{(l)} + \left(1 - \widehat{\rho}^{(l)}\right) e^{-\widehat{\lambda}^{(l)}}\right\} = 0.6898,$$

where $\widehat{\rho}^{(l)}$ and $\widehat{\lambda}^{(l)}$ are the sample draw of the posterior distribution of ρ and λ, respectively, obtained at the l—th iteration of the MCMC algorithm.

As recommended by Lin and Lee [21], in the context of dependent longitudinal data, a more appropriate measure of fitness is the predictive accuracy of future observations. To obtain two-step-ahead predictions of the last two values of the data (observations 421 and 422), we employ Step 5 of the algorithm in Section 4.2, under the ZINAR(3) model.

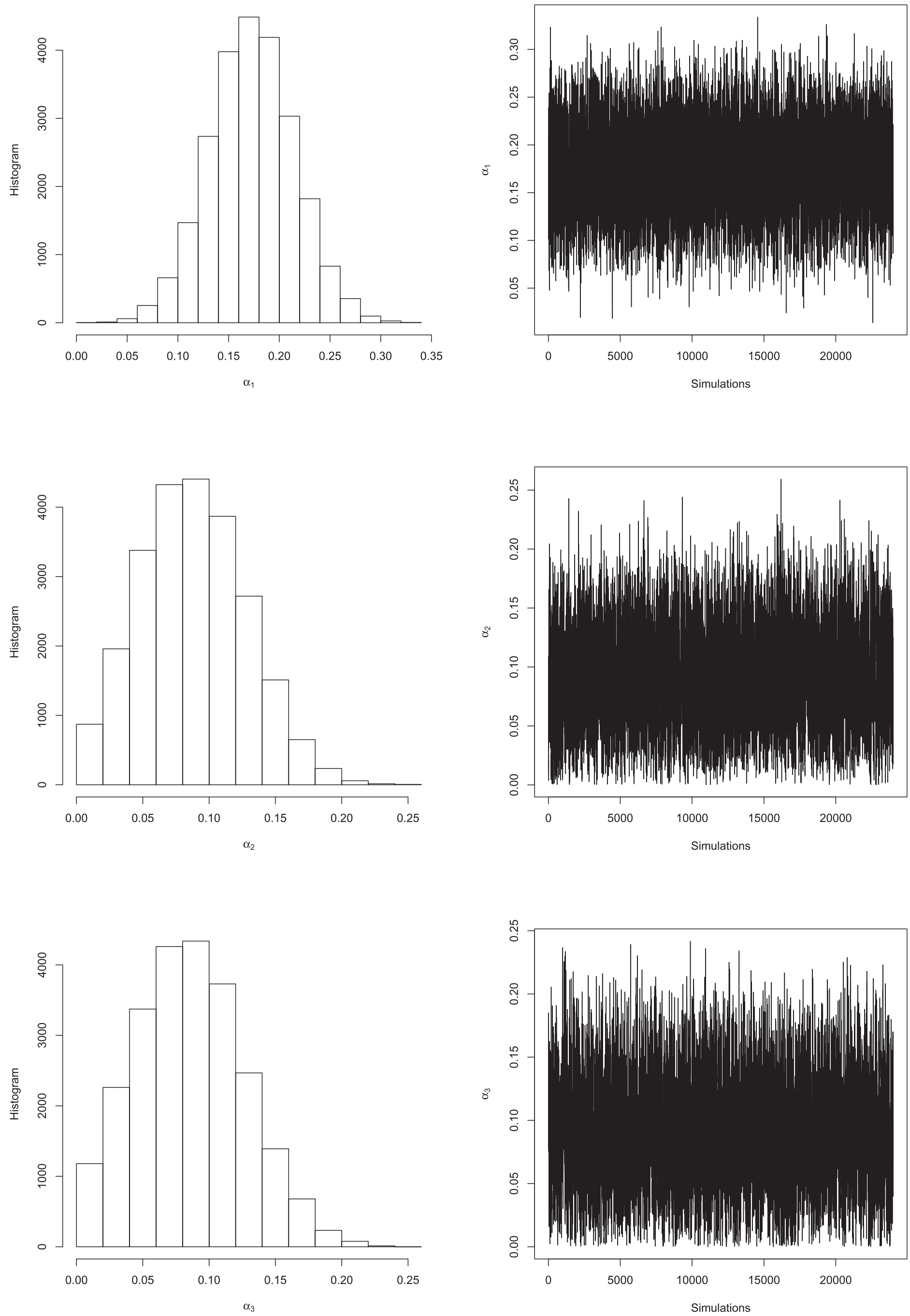

Figure 3. Assaults data set. History of the chains and the approximate posterior marginal densities of α_1, α_2 and α_3 considering the ZINAR(3) process.

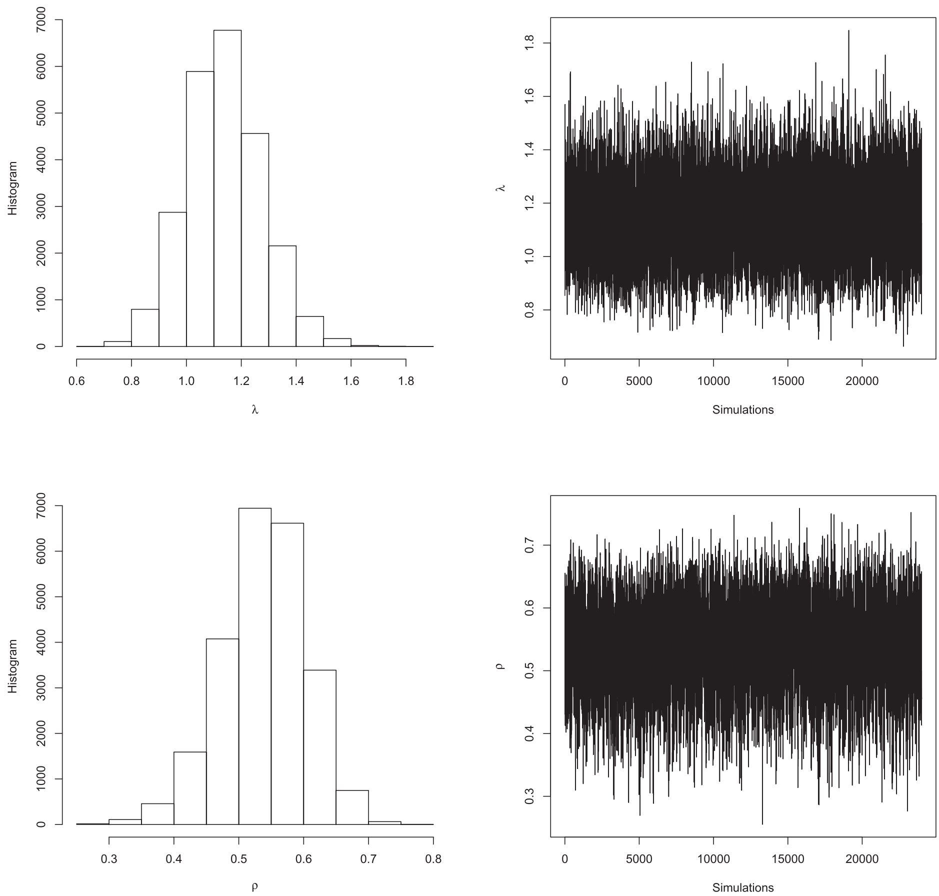

Figure 4. Assaults data set. History of the chains and the approximate posterior marginal densities of λ and ρ under the ZINAR(3) process.

Figure 2 depicts the bar plots of the predictive distributions of the data points 421 and 422, showing a close agreement with the bar plot of the data given in Figure 1.

As an additional information, Figures 3 and 4 depict the history of the MCMC chains and the approximate posterior marginal densities of the parameters under the ZINAR(3) model.

6. Conclusions

This article has introduced the ZINAR(p) process, which is an extension of the ZINAR(1) process, with order p bigger than one. The aim is to propose a model with more flexibility and, beside this, obtain a process that deals with excess of zeros. For fitting the proposed model, we develop an efficient MCMC algorithm for conducting Bayesian inference and its performance was evaluated through simulations and a real data from a strategic security agency of the state of Pernambuco, Brazil. In this illustrative example, there is strong evidence of high-order dependence and inflated zero counts. The implementation of the algorithm was done by using the R package [25] and the source code is available on request from the corresponding author. We conjecture that our methods can be extended by adding a moving average structure to the process. An in-depth investigation of such an extension is beyond the scope of the present paper, but deserves an interesting topic for further research.

Disclosure statement

No potential conflict of interest was reported by the authors.

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Appendices

Appendix 1. Proof of Equation (21)

A.1 The distribution of $Z, S, W, \theta | y$

Recalling that $\boldsymbol{\theta} = (\boldsymbol{\alpha}, \rho, \lambda)$, we consider the partition of the sample $\mathbf{y} = \{y_1, y_2, \ldots, y_{n+p}\}$ into the sub samples $\mathbf{y}_1 = \{y_1, \ldots, y_p\}$ and $\mathbf{y}_2 = \{y_{p+1}, \ldots, y_{p+n}\}$. Then,

$$\pi\left(\mathbf{z}, \mathbf{s}, \mathbf{w}, \boldsymbol{\theta} | \mathbf{y}\right) \propto \pi\left(\mathbf{z}, \mathbf{s}, \mathbf{w}, \boldsymbol{\theta}, \mathbf{y}_2 | \mathbf{y}_1\right) \propto \pi\left(\mathbf{z}, \mathbf{s}, \mathbf{w}, \mathbf{y}_2 | \boldsymbol{\theta}, \mathbf{y}_1\right) \pi(\mathbf{y}_1 | \boldsymbol{\theta}) \pi(\boldsymbol{\theta}). \tag{A1}$$

As described in Section 3.3, the likelihood $\pi(\mathbf{y}_1|\boldsymbol{\theta})$ is unknown in general, and a usual procedure is to discard the contribution of $\mathbf{y}_1$ and carry out the conditional estimation method.

For $t = p+1, \ldots, p+n$, let us define $\boldsymbol{\xi}_t = (z_t, \mathbf{s}_t, w_t)$. Recalling that $\mathbf{y}_{(t-p)} = \{y_{t-p}, \ldots, y_{t-1}\}$, we get

$$\begin{aligned}
\pi\left(\mathbf{z}, \mathbf{s}, \mathbf{w}, \mathbf{y}_2 | \boldsymbol{\theta}, \mathbf{y}_1\right) &= \pi(\boldsymbol{\xi}_{p+1}, y_{p+1} | \mathbf{y}_1, \boldsymbol{\theta}) \prod_{t=p+2}^{p+n} \pi(\boldsymbol{\xi}_t, y_t | \boldsymbol{\xi}_{p+1}, \ldots, \boldsymbol{\xi}_{t-1}, \mathbf{y}_1, y_{p+1}, \ldots, y_{t-1}, \boldsymbol{\theta}) \\
&= \pi(\boldsymbol{\xi}_{p+1}, y_{p+1} | \mathbf{y}_1, \boldsymbol{\theta}) \\
&\quad \times \prod_{t=p+2}^{p+n} \pi(\boldsymbol{\xi}_t, y_t | y_1, \ldots, y_{t-p-1}, \mathbf{y}_{(t-p)}, \boldsymbol{\theta})
\end{aligned} \tag{A2}$$

$$
\begin{aligned}
&= \pi(\boldsymbol{\xi}_{p+1}, y_{p+1} | \mathbf{y}_1, \boldsymbol{\theta}) \\
&\quad \times \prod_{t=p+2}^{p+n} \pi(\boldsymbol{\xi}_t, y_t | \mathbf{y}_{(t-p)}, \boldsymbol{\theta}),
\end{aligned} \tag{A3}
$$

where, to obtain (A2), we used assumptions 2.1 and 2.1 in Definition 2.1 and, to obtain (A3), the fact that given $\mathbf{y}_{(t-p)}$, $(\boldsymbol{\xi}_t, y_t)$ and $(y_1, \ldots, y_{t-p-1})$ are independent.

Now observe that, for $t = p+1, \ldots, p+n$,

$$
\begin{aligned}
\pi(\boldsymbol{\xi}_t, y_t |\, \mathbf{y}_{(t-p)}, \boldsymbol{\theta}) &= \pi(y_t | \boldsymbol{\xi}_t, \mathbf{y}_{(t-p)}, \boldsymbol{\theta}) \pi(\boldsymbol{\xi}_t | \mathbf{y}_{(t-p)}, \boldsymbol{\theta}) \\
&= \mathbb{I}_{\{z_t + \sum_{i=1}^{p} s_{t,i}\}}(y_t) \pi(\boldsymbol{\xi}_t | \mathbf{y}_{(t-p)}, \boldsymbol{\theta}) \\
&\propto \pi\left(\boldsymbol{\xi}_t | y_t, \mathbf{y}_{(t-p)}, \boldsymbol{\theta}\right),
\end{aligned}
$$

see Equations (19) and (20). Using this result in (A3), we get

$$
\pi\left(\mathbf{z}, \mathbf{s}, \mathbf{w}, \mathbf{y}_2 | \boldsymbol{\theta}, \mathbf{y}_1\right) \propto \prod_{t=p+1}^{p+n} \pi\left(\boldsymbol{\xi}_t | y_t, \mathbf{y}_{(t-p)}, \boldsymbol{\theta}\right).
$$

Discarding $\pi(\mathbf{y}_1 | \boldsymbol{\theta})$ in (A1) gives

$$
\pi\left(\mathbf{z}, \mathbf{s}, \mathbf{w}, \boldsymbol{\theta} | \mathbf{y}\right) \propto \pi(\boldsymbol{\theta}) \prod_{t=p+1}^{p+n} \pi\left(\boldsymbol{\xi}_t | y_t, \mathbf{y}_{(t-p)}, \boldsymbol{\theta}\right),
$$

which is Equation (21).

Appendix 2. Results of Artificial data set (Section 5.1)

Table A1. Artificial data sets. Summary of MCMC results based on 1000 simulated samples from the ZINAR(p) process, for $\rho = 0.40$, $n = 80$ and $p = 1, \ldots, 4$.

			MCMC summary		
Model	Parameter	Real value	MC Mean	MC SD	MC Cov
	α	**0.40**	0.3811	0.0790	94.9%
ZINAR(1)	λ	**2.00**	1.9805	0.3049	94.8%
	ρ	**0.40**	0.3653	0.1032	94.0%
	α_1	**0.40**	0.3711	0.0910	96.2%
ZINAR(2)	α_2	**0.20**	0.2052	0.0879	96.6%
	λ	**2.00**	2.0267	0.4030	96.4%
	ρ	**0.40**	0.3558	0.1384	96.7%
	α_1	**0.20**	0.2050	0.0874	96.4%
	α_2	**0.30**	0.2671	0.0908	95.4%
ZINAR(3)	α_3	**0.15**	0.1740	0.0831	96.6%
	λ	**2.00**	1.9811	0.5141	94.0%
	ρ	**0.40**	0.3642	0.1677	99.1%
	α_1	**0.30**	0.2739	0.0893	95.9%
	α_2	**0.20**	0.1984	0.0858	97.7%
ZINAR(4)	α_3	**0.15**	0.1725	0.0810	96.2%
	α_4	**0.10**	0.1115	0.0737	93.2%
	λ	**2.00**	1.9718	0.2045	98.0%
	ρ	**0.40**	0.3662	0.2144	97.6%

Table A2. Artificial data sets. Summary of MCMC results based on 1000 simulated samples from the ZINAR(p) process, for $\rho = 0.40$, $n = 150$ and $p = 1, \ldots, 4$.

			MCMC summary		
Model	Parameter	Real value	MC Mean	MC SD	MC Cov
	α	**0.40**	0.3885	0.0575	95.6%
ZINAR(1)	λ	**2.00**	1.9843	0.2238	93.6%
	ρ	**0.40**	0.3789	0.0772	93.1%
	α_1	**0.40**	0.3830	0.0681	95.3%
ZINAR(2)	α_2	**0.20**	0.1963	0.0695	96.0%
	λ	**2.00**	2.0173	0.2941	95.8%
	ρ	**0.40**	0.3665	0.1089	95.3%
	α_1	**0.20**	0.1942	0.0697	94.7%
	α_2	**0.30**	0.2801	0.0706	94.5%
ZINAR(3)	α_3	**0.15**	0.1602	0.0675	97.9%
	λ	**2.00**	2.0166	0.3512	94.6%
	ρ	**0.40**	0.3707	0.1307	96.7%
	α_1	**0.30**	0.2774	0.0735	94.5%
	α_2	**0.20**	0.1875	0.0726	95.9%
ZINAR(4)	α_3	**0.15**	0.1562	0.0686	98.3%
	α_4	**0.10**	0.1184	0.0615	98.2%
	λ	**2.00**	2.0245	0.4908	93.5%
	ρ	**0.40**	0.3709	0.1625	98.3%

Table A3. Artificial data sets. Summary of MCMC results based on 1000 simulated samples from the ZINAR(p) process, for $\rho = 0.40$, $n = 300$ and $p = 1, \ldots, 4$.

			MCMC summary		
Model	Parameter	Real value	MC Mean	MC SD	MC Cov
	α	**0.40**	0.3952	0.0401	95.0%
ZINAR(1)	λ	**2.00**	1.9955	0.1575	94.0%
	ρ	**0.40**	0.3911	0.0542	94.2%
	α_1	**0.40**	0.3912	0.0486	96.5%
ZINAR(2)	α_2	**0.20**	0.1966	0.0521	95.1%
	λ	**2.00**	1.9999	0.2079	94.4%
	ρ	**0.40**	0.3758	0.0807	94.7%
	α_1	**0.20**	0.1951	0.0529	92.8%
	α_2	**0.30**	0.2931	0.0515	95.0%
ZINAR(3)	α_3	**0.15**	0.1497	0.0519	94.3%
	λ	**2.00**	2.0097	0.2525	96.1%
	ρ	**0.40**	0.3858	0.1000	94.0%
	α_1	**0.30**	0.2912	0.0537	95.2%
	α_2	**0.20**	0.1933	0.0559	95.8%
ZINAR(4)	α_3	**0.15**	0.1474	0.0535	96.9%
	α_4	**0.10**	0.1086	0.0486	95.8%
	λ	**2.00**	2.0153	0.3195	95.4%
	ρ	**0.40**	0.3775	0.1259	97.3%

Table A4. Artificial data sets. Summary of MCMC results based on 1000 simulated samples from the ZINAR(p) process, for $\rho = 0.70$, $n = 80$ and $p = 1, \ldots, 4$.

			MCMC summary		
Model	Parameter	Real value	MC Mean	MC SD	MC Cov
	α	**0.40**	0.3834	0.0754	95.7%
ZINAR(1)	λ	**2.00**	1.9021	0.3990	92.7%
	ρ	**0.70**	0.6575	0.0846	93.2%
	α_1	**0.40**	0.3744	0.0858	94.8%
ZINAR(2)	α_2	**0.20**	0.2004	0.0812	95.7%
	λ	**2.00**	1.8883	0.4536	90.7%
	ρ	**0.70**	0.6184	0.1134	91.8%
	α_1	**0.20**	0.1990	0.0822	95.6%
	α_2	**0.30**	0.2669	0.0865	95.2%
ZINAR(3)	α_3	**0.15**	0.1628	0.0780	97.5%
	λ	**2.00**	1.8752	0.5038	98.5%
	ρ	**0.70**	0.6065	0.1417	91.6%
	α_1	**0.30**	0.2656	0.0898	95.0%
	α_2	**0.20**	0.1835	0.0844	97.6%
ZINAR(4)	α_3	**0.15**	0.1579	0.0789	98.1%
	α_4	**0.10**	0.1282	0.0713	96.2%
	λ	**2.00**	1.8923	0.6245	96.0%
	ρ	**0.70**	0.6345	0.1764	92.7%

Table A5. Artificial data sets. Summary of MCMC results based on 1000 simulated samples from the ZINAR(p) process, for $\rho = 0.70$, $n = 150$ and $p = 1, \ldots, 4$.

			MCMC summary		
Model	Parameter	Real value	MC Mean	MC SD	MC Cov
	α	**0.40**	0.3903	0.0546	95.4%
ZINAR(1)	λ	**2.00**	1.9617	0.2903	93.9%
	ρ	**0.70**	0.6816	0.0563	95.0%
	α_1	**0.40**	0.3875	0.0632	95.5%
ZINAR(2)	α_2	**0.20**	0.1984	0.0635	95.2%
	λ	**2.00**	1.9312	0.3392	92.6%
	ρ	**0.70**	0.6557	0.0782	93.5%
	α_1	**0.20**	0.1994	0.0656	94.7%
	α_2	**0.30**	0.2799	0.0664	96.0%
ZINAR(3)	α_3	**0.15**	0.1542	0.0629	96.9%
	λ	**2.00**	1.9904	0.3794	92.3%
	ρ	**0.70**	0.6543	0.0978	93.8%
	α_1	**0.30**	0.2784	0.0703	94.4%
	α_2	**0.20**	0.1827	0.0695	96.2%
ZINAR(4)	α_3	**0.15**	0.1504	0.0656	97.2%
	α_4	**0.10**	0.1150	0.0586	97.1%
	λ	**2.00**	1.9093	0.4370	90.1%
	ρ	**0.70**	0.6654	0.1292	90.7%

Table A6. Artificial data sets. Summary of MCMC results based on 1000 simulated samples from the ZINAR(p) process, for $\rho = 0.70$, $n = 300$ and $p = 1, \ldots, 4$.

			MCMC summary		
Model	Parameter	Real value	MC Mean	MC SD	MC Cov
	α	**0.40**	0.3952	0.0382	94.8%
ZINAR(1)	λ	**2.00**	1.9903	0.2085	94.1%
	ρ	**0.70**	0.6900	0.0395	94.8%
	α_1	**0.40**	0.3932	0.0446	94.6%
ZINAR(2)	α_2	**0.20**	0.1976	0.0458	95.0%
	λ	**2.00**	1.9739	0.2353	94.4%
	ρ	**0.70**	0.6817	0.0500	94.7%
	α_1	**0.20**	0.1934	0.0468	95.2%
	α_2	**0.30**	0.2804	0.0457	94.2%
ZINAR(3)	α_3	**0.15**	0.1411	0.0460	92.2%
	λ	**2.00**	1.9966	0.2596	91.5%
	ρ	**0.70**	0.6854	0.06241	90.1%
	α_1	**0.30**	0.2903	0.0506	96.2%
	α_2	**0.20**	0.1897	0.0528	94.7%
ZINAR(4)	α_3	**0.15**	0.1462	0.0508	94.0%
	α_4	**0.10**	0.1032	0.0456	95.7%
	λ	**2.00**	1.9476	0.3149	93.2%
	ρ	**0.70**	0.6406	0.0838	92.8%